

Introduction to quantum computing, basics of Quantum Algorithms Using Qiskit I

Dr Dauda Alani Dikko

Department of Mathematics
Faculty of Science
University of Ibadan
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1. Introduction & Background
2. Some Mathematical Framework
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Introduction to Quantum Computation

Quantum computation exploits quantum mechanics principles for information processing, fundamentally different from classical computation.

Quantum computation makes direct use of quantum mechanics phenomena such as superposition and entanglement to perform operations on data.

The basic principle can be used to represent data and perform operations on these data (Vlatko Vedral, 1998).

In quantum computation, the state space of a quantum system is described by a Hilbert space.

A Hilbert space $\mathcal{H}$ is a complete complex inner product space. In quantum computation, the state space of a quantum system is described by a Hilbert space (Nielsen & Chuang, 2010).

For a single qubit, the Hilbert space is $\mathcal{H} = \mathbb{C}^2$. For an n -qubit system, the space becomes $\mathcal{H}^{\otimes n} \cong \mathbb{C}^{2^n}$.

- ▶ Quantum computation uses quantum phenomena (superposition, entanglement) to process data.
- ▶ Quantum algorithms are probabilistic, offering high probability of correct solutions.
- ▶ Key principles: quantum parallelism, interference, and unitary transformations.
- ▶ Major algorithms:
 1. Shor's Factoring Algorithm
 2. Grover's Search Algorithm
 3. Simon's Algorithm
 4. Deutsch-Jozsa Algorithm

Classical Computing

- ▶ Bits: 0 or 1
- ▶ Deterministic operations
- ▶ Sequential processing
- ▶ No superposition

Quantum Computing

- ▶ Qubits: $\alpha|0\rangle + \beta|1\rangle$
- ▶ Superposition states
- ▶ Quantum parallelism
- ▶ Entanglement

Unitary Operators

An operator $U : \mathcal{H} \rightarrow \mathcal{H}$ is unitary if $U^\dagger U = I$, where $U^\dagger$ is the conjugate transpose. Unitary operators preserve the inner product and describe reversible quantum evolution

Quantum Gates and Circuits

Basic Quantum Gates

- The Pauli matrices are:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- ▶ The Hadamard gate is defined as:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It creates superposition: $H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

- ▶ The Controlled-NOT gate is a two-qubit entangling gate:

$$\text{CNOT} |a, b\rangle = |a, a \oplus b\rangle.$$

Measurement and Observeables

- ▶ A projective measurement is described by a set of orthogonal projectors $\{P_i\}$ satisfying $\sum_i P_i = I$. The probability of outcome i given state $|\psi\rangle$ is:

$$\Pr(i) = \langle \psi | P_i | \psi \rangle.$$

- ▶ (Born Rule) If $|\psi\rangle$ is measured in basis $\{|i\rangle\}$, the probability of outcome i is $|\langle i | \psi \rangle|^2$ (Nielsen & Chuang, 2010).
- ▶ (Entangled State) A pure state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is *entangled* if it cannot be written as a tensor product $|\psi_A\rangle \otimes |\psi_B\rangle$. Example: the Bell state $|\blacksquare^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ (Nielsen & Chuang, 2010).

Quantum Fourier Transform

For $N = 2^n$, the QFT on an n -qubit state is:

$$\text{QFT}|x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{2\pi i xy/N} |y\rangle.$$

It can be implemented with $O(n^2)$ gates (Nielsen & Chuang, 2010).

Oracle Model

(Quantum Oracle) Given a function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$, a quantum oracle U_f is a unitary operator such that:

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle.$$

This model is used in Deutsch–Jozsa, Simon’s, and Grover’s algorithms (Nielsen & Chuang, 2010).

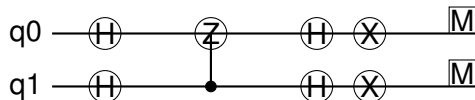
Key References

- ▶ Nielsen & Chuang (2010): Quantum information theory
- ▶ Watrous (2018): Mathematical foundations
- ▶ Montanaro (2016): Quantum algorithms survey

Implementation Frameworks

- ▶ Qiskit (IBM) - Used in this project
- ▶ Cirq (Google)
- ▶ ProjectQ (ETH Zurich)

Grover's Algorithm Circuit (2 qubits)



Circuit Components

H: Hadamard (superposition), CZ: Oracle (marks $|11\rangle$), X: Bit flip, M: Measurement

Definition

The Quantum Fourier Transform is a linear transformation on quantum bits, analogous to the discrete Fourier transform.

- ▶ Crucial for Shor's algorithm and period-finding.
- ▶ For $Q = 2^q$, QFT maps:

$$|x\rangle \rightarrow \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} \omega^{xy} |y\rangle, \quad \omega = e^{2\pi i/Q}$$

- ▶ Enables interference patterns to reveal periodicities.

Purpose

Factorizes integers in polynomial time, threatening RSA encryption.

- ▶ **Classical part:** Reduces factoring to order-finding.
- ▶ **Quantum part:** Uses QFT to find period r of $f(x) = a^x \bmod N$.
- ▶ Complexity: $O((\log N)^5)$ (quantum) vs. exponential (classical).

Purpose

Searches an unsorted database of N entries in $O(\sqrt{N})$ time.

- ▶ Quadratic speedup over classical $O(N)$.
- ▶ Uses amplitude amplification via operators:

$$U_{\omega} = I - 2|\omega\rangle\langle\omega|, \quad U_s = 2|s\rangle\langle s| - I$$

- ▶ Generalizes to multiple search targets: $O(\sqrt{N/L})$ for L matches.

Problem

Determine if $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is constant or balanced.

- ▶ Classical worst case: $2^{n-1} + 1$ queries.
- ▶ Quantum: always correct with **one** query.
- ▶ Uses superposition and interference:

$$|0\rangle^{\otimes n} |1\rangle \xrightarrow{H} \frac{1}{\sqrt{2^{n+1}}} \sum_x |x\rangle (|0\rangle - |1\rangle)$$

- ▶ Final measurement yields $|0\rangle^{\otimes n}$ iff f is constant.

Problem

Given $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ with $f(x) = f(y) \iff y = x \oplus s$, find s .

- ▶ Provides exponential speedup: $O(n)$ queries vs. $\blacksquare(2^{n/2})$ classically.
- ▶ Inspired Shor's algorithm.
- ▶ Uses Hadamard transforms and quantum parallelism.

Simulation Results Summary

Algorithm	Qubits	Shots	Success	Depth
Deutsch	2	1024	100%	4
Deutsch-Jozsa	4	1024	100%	6
Grover	2	1024	94.5%	12
QFT	3	1024	100%	9
Shor (N=15)	8	2048	92%	120

Grover's Algorithm Results

Measurement Statistics (1024 shots)

State	Counts	Percentage
$ 11\rangle$ (marked)	945	92.3%
$ 00\rangle$	26	2.5%
$ 01\rangle$	27	2.6%
$ 10\rangle$	26	2.5%

Analysis

Theoretical probability: 94.5%

Empirical result: 92.3%

Small deviation due to finite sampling

Algorithm	Classical	Quantum
Deutsch	2 queries	1 query
Deutsch-Jozsa	2^{n-1} queries	1 query
Grover	$O(N)$ queries	$O(\sqrt{N})$ queries
Shor	Exponential	$O(n^3)$ gates
QFT	$O(N^2)$ ops	$O(n^2)$ gates

Technical Limitations

- ▶ Simulation memory: $O(2^n)$ requirements
- ▶ Gate decomposition overhead
- ▶ Circuit depth limitations
- ▶ Measurement noise effects

Mitigation Strategies

- ▶ Optimized circuit decompositions
- ▶ Error mitigation techniques
- ▶ Quantum volume benchmarking
- ▶ Circuit transpilation

Key Achievements

- ▶ Established mathematical framework for quantum algorithms
- ▶ Successfully implemented algorithms in Qiskit
- ▶ Verified theoretical speedups through simulation
- ▶ Grover's algorithm: 92.3% success rate
- ▶ Demonstrated practical quantum advantage

Research Directions

- ▶ Noise-robust algorithm implementations
- ▶ Hybrid quantum-classical approaches
- ▶ Domain-specific adaptations
- ▶ NISQ device optimization

Educational Value

- ▶ Project serves as learning resource
- ▶ Qiskit implementations for education
- ▶ Demonstrates practical implementation

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Thank You!